In-Plane Orthotropic Elastic Wave Control by Fiber-Orientation Design

model, adjoint, algorithms, results

July 20, 2026

The question

Can we steer elastic waves in a composite plate by designing *only* the local fiber orientation $\theta(\mathbf{x})$ — no topology, one material?

Why it might work. A continuous-fiber composite is strongly orthotropic:

$$E_{f1}/E_{f2} \approx 15 \implies c_{\text{fast}}/c_{\text{slow}} \approx 3.8$$

Why it might not. Orientation is a single scalar per point, and the wave equation is only weakly coupled to it … unless the material is anisotropic.

The lever, precisely

$$\mathbf{D}_f(\theta) = \mathbf{T}(\theta)\,\mathbf{Q}_f\,\mathbf{T}(\theta)^{\mathsf{T}}$$

$$\mathbf{Q}_f \text{ isotropic} \iff \frac{\partial \mathbf{D}_f}{\partial \theta} \equiv \mathbf{0}$$

Every gradient in this work is built from $\partial \mathbf{D}_f/\partial\theta$, so for an isotropic plate the design sensitivity is *identically zero*.

Model

Time-harmonic in-plane elastodynamics

Plane stress, $\mathbf{u}(\mathbf{x})e^{-\mathrm{i}\omega t}$, rotated orthotropic stiffness:

$$\big[(1+\mathrm{i}\eta)\mathbf{K}(\boldsymbol{\theta}) - \omega^2\mathbf{M} + \mathrm{i}\omega\mathbf{C}\big]\mathbf{u} = \mathbf{f}$$

- $\mathbf{K}$ from $\mathbf{D}_f(\theta)$ — the only θ-dependent operator
- η: structural (hysteretic) damping; $\mathbf{C}$: quadratic absorbing sponge
- genuine *vector* physics, not a reduced antiplane scalar model

The installed PETSc is a real build
So the complex system is solved by an exact real four-field split — and the split has to be provably lossless, not merely convenient.

The real four-field split

With $A = \mathbf{K} - \omega^2 \mathbf{M}$ and $B = \eta \mathbf{K} + \omega \mathbf{C}$ (both real symmetric),

$$(A + \mathrm{i}B)(\mathbf{u}_r + \mathrm{i}\mathbf{u}_i) = \mathbf{f} \iff \underbrace{\begin{bmatrix} A & -B \\ B & A \end{bmatrix}}_{\mathcal{A}} \begin{bmatrix} \mathbf{u}_r \\ \mathbf{u}_i \end{bmatrix} = \begin{bmatrix} \mathbf{f}_r \\ \mathbf{f}_i \end{bmatrix}$$

Two facts that make this free

1. Equating real and imaginary parts is exact ($\{1, \mathrm{i}\}$ is a basis of $\mathbb{C}$ over $\mathbb{R}$) — nothing is approximated.
2. $\mathcal{A}^{\mathsf{T}}$ is the realification of $(A + \mathrm{i}B)^{\mathsf{H}}$.
   So the transposed real solve returns the true complex adjoint.

Cost: $\sim 2\times$ a native complex LU. With complex `PETSc` the whole construction collapses to one line, and no reported number changes.

Discrete spaces

	wave control (`dolfinx`)	periodic cell (Q4, `numpy/scipy`)
cells	triangles ($\mathbb{P}_1$)	rectangles (Q4 bilinear)
state	$[\mathbb{P}_1]^4$, 4 dof/node	Q4 vector, 2 dof/node
	$(u_x^r, u_y^r, u_x^i, u_y^i)$	(u_x, u_y)
design	$\mathbb{DG}_0$ per cell	element-wise constant
solve	sparse LU (MUMPS)	dense Hermitian `eigh`
dual	adjoint in the same space, $\mathcal{A}^\mathsf{T}$	eigenvalue perturbation

- Holes are *cut from the geometry* (conforming `gmsh`), not soft SIMP voids — a staircased rim scatters on its own account and blurs the very radius the curvature rule is tied to.
- Two independent codes share the physics and the design machinery, but no assembly or linear algebra: each is a check on the other.

Sensitivity and algorithms

Discrete adjoint in two lines of UFL

Adjoin $\mathcal{A}(\boldsymbol{\theta})\mathsf{U} = \mathsf{b}$ with $\boldsymbol{\Lambda}$; kill the implicit state term:

$$\boxed{\mathcal{A}^{\mathsf{T}}\boldsymbol{\Lambda} = -\partial J/\partial \mathsf{U}} \qquad \boxed{\frac{\mathrm{d}J}{\mathrm{d}\theta_\ell} = \frac{\partial R}{\partial \theta_\ell}}$$

$$\frac{\mathrm{d}J}{\mathrm{d}\theta_\ell} = \int_{\Omega_\ell} \left[\boldsymbol{\varepsilon}(\mathbf{u}_r)^{\mathsf{T}} \tfrac{\partial \mathbf{D}_f}{\partial \theta} \boldsymbol{\varepsilon}(\boldsymbol{\lambda}_r) + \boldsymbol{\varepsilon}(\mathbf{u}_i)^{\mathsf{T}} \tfrac{\partial \mathbf{D}_f}{\partial \theta} \boldsymbol{\varepsilon}(\boldsymbol{\lambda}_i) + \dots \right]$$

- mass, sponge and source carry no θ: they drop out
- the whole gradient is built from $\partial \mathbf{D}_f/\partial\theta$ — hence the isotropy result is exact, not asymptotic
- FD-verified to $\sim 10^{-6}$; one LU reused for forward *and* adjoint

Two gradient chains

Chain A — orientation-only wave control

$$\hat{\theta} \xrightarrow{\mathbf{S}} \hat{\theta}^{\mathrm{sym}} \xrightarrow{\mathbf{B}} \theta \xrightarrow{\mathbf{D}_f} \mathcal{A} \xrightarrow{\mathrm{solve}} \mathsf{U} \longrightarrow J$$

$$\nabla_{\hat{\theta}}\big[-\ln E + \mu\Pi\big] = -\tfrac{1}{E}\,\mathbf{S}^{\mathsf{T}}\mathbf{B}^{\mathsf{T}}\mathbf{d} + \mu\,\mathbf{S}^{\mathsf{T}}(\partial\zeta/\partial\hat{\theta})^{\mathsf{T}}\mathbf{q}$$

Chain B — two-field Bloch co-design

$$z \xrightarrow{\mathbf{P}} \tilde{y} \xrightarrow{\mathbf{E}_{\mathcal{F}}} y \xrightarrow{\mathrm{SIMP}} \mathbf{K}, \mathbf{M}, \qquad \hat{\theta} \xrightarrow{\mathbf{B}} \theta$$

$$\nabla_z J = \mathbf{P}^{\mathsf{T}}\mathbf{E}_{\mathcal{F}}\Big[\partial J/\partial y + 2c_v(\bar{z} - v^\star)\mathbf{A}/\textstyle\sum_j A_j\Big]$$

Every operator contributes exactly one transposed factor. $\mathbf{S}$ is an orthogonal projector, so symmetrizing the design and the gradient are the *same* operation.

Manufacturing enters the optimization, not the post-processing

CS-RBF orientation

$\theta = \mathbf{B}\hat{\theta}$, Wendland C^2, row-stochastic

Fiber curvature

$$\zeta = (\mathbf{n}\cdot\nabla)\theta, \qquad |\zeta| \le \zeta_{\text{all}}$$

$\zeta_{\text{all}} = 1/R$ for a hole the tows must wrap; $1/R_{\text{tow}}$ for a steered channel — *these are different constraints*.

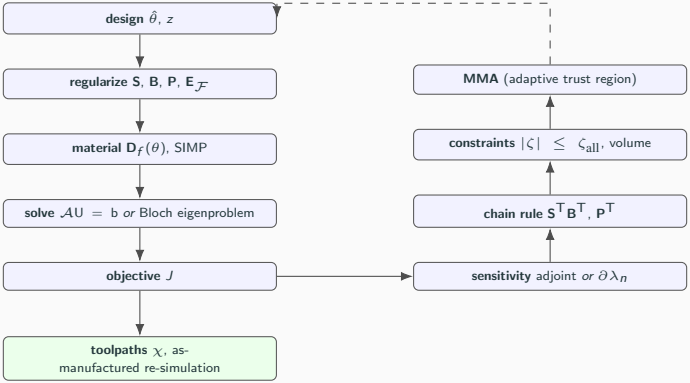
Toolpaths by wave projection

$$\nabla\psi = [-\sin\theta, \cos\theta]^{\mathsf{T}}, \quad \chi = \tfrac{1}{2} + \tfrac{1}{2}\cos(2\pi\psi/d)$$

Solved as a least-squares gradient match — the normal-equations Poisson form drops the boundary flux and collapses the pitch.

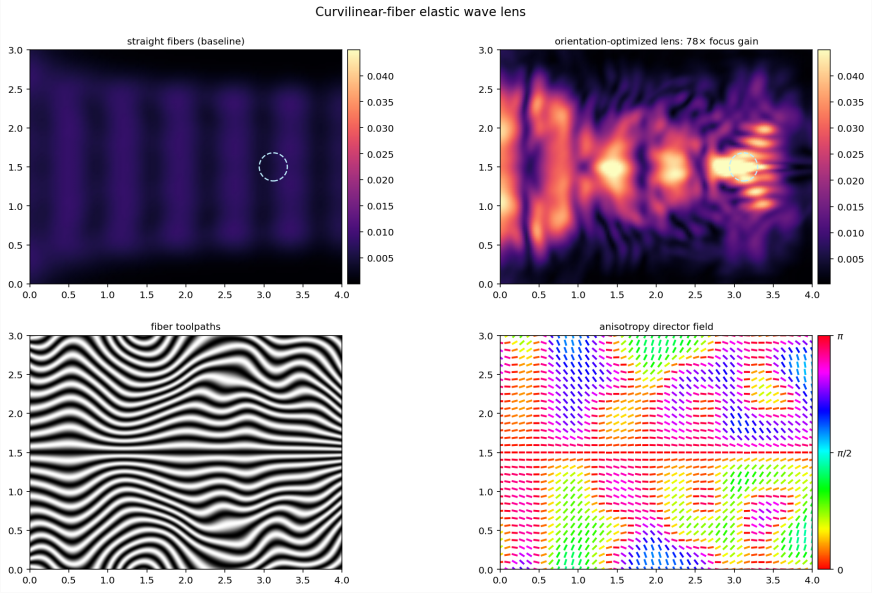
On a periodic cell: FFT Helmholtz projection + an *integer* number of tow cycles, so tows join across every boundary.

Data flow

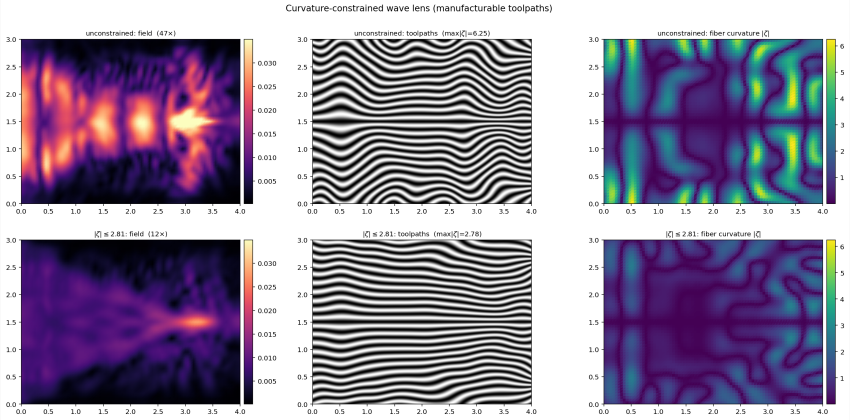

design $\hat{\theta}$, z
regularize $\mathbf{S}$, $\mathbf{B}$, $\mathbf{P}$, $\mathbf{E}_{\mathcal{F}}$
material $\mathbf{D}_f(\theta)$, SIMP
solve $\mathcal{A}\mathbf{U} = \mathbf{b}$ *or* Bloch eigenproblem
objective J
toolpaths χ, as-manufactured re-simulation
MMA (adaptive trust region)
constraints $|\zeta| \leq \zeta_{\mathrm{all}}$, volume
chain rule $\mathbf{S}^{\mathsf{T}}\mathbf{B}^{\mathsf{T}}$, $\mathbf{P}^{\mathsf{T}}$
sensitivity adjoint *or* $\partial\lambda_n$

Results

Curvilinear-fiber wave lens

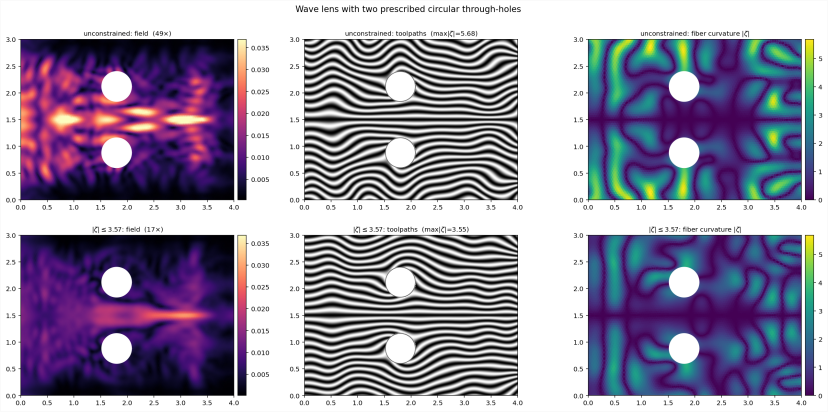


Manufacturability costs performance — honestly



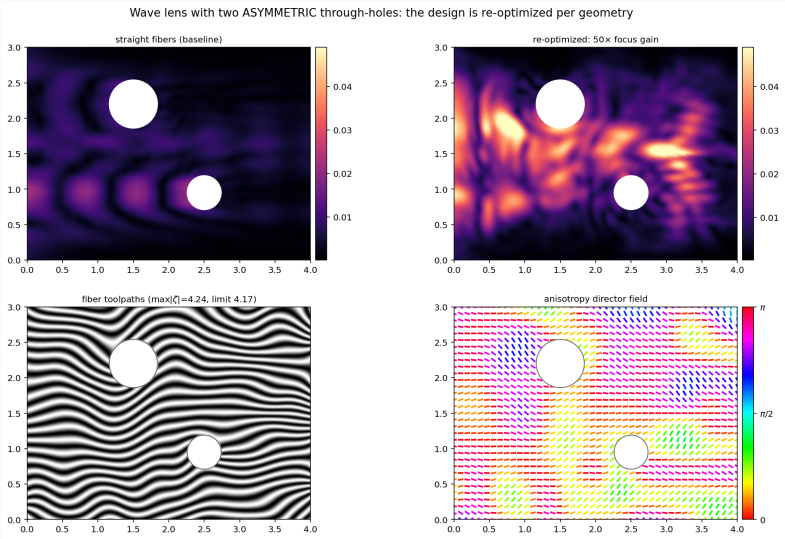
Bounding $|\zeta| \leq 2.8$ relaxes the tight bends and the gain falls $47\times \to 12\times$. This trade-off is the point, not a defect.

Designing around prescribed through-holes



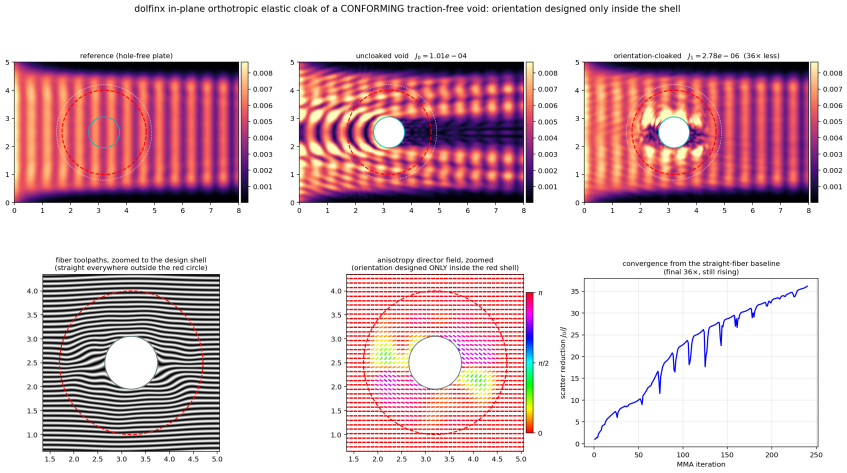
Holes *cut from the geometry*: 49× unconstrained, 17× under $|\zeta| \leq 1/R = 3.57$ (feasible: $3.55 < 3.57$).

The design is re-solved per geometry



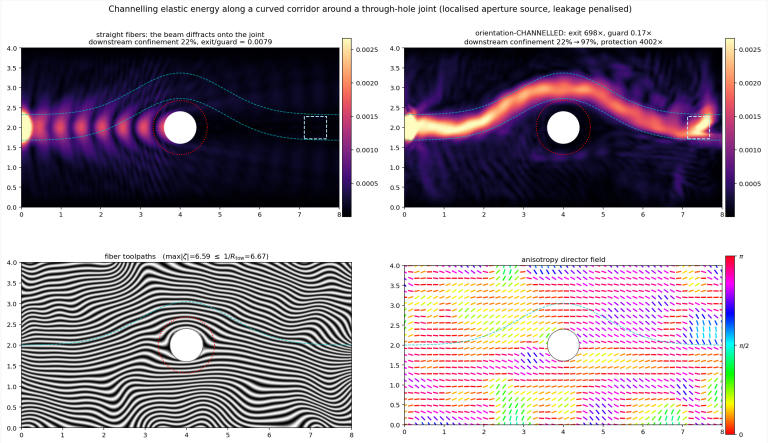
Asymmetric holes, symmetry constraint dropped: 50×. Caveat: converges to $|\zeta|_{\max} = 4.24$ against a limit of 4.17 — a 1.7% overshoot. The penalty is smooth, not a hard constraint.

Elastic cloak of a conforming void



Orientation designed *only* inside the red shell (verified $\max|\theta| = 0$ outside): **36**× scatter reduction, still rising at 240 iterations. Soft-void posing of the same problem gives markedly less.

Routing energy around a joint



Exit 698×, guard energy 0.17 of baseline, downstream confinement 22% → 97%, feasible at $|\zeta|_{\max} = 6.59 \leq 6.67$.

What made the routing problem work

Three modelling choices, each found by a *silent* failure:

1. **Localized source.** A plane wave leaves most of the field outside the corridor — nothing to channel.
2. **Print-head curvature limit, not the hole's.** $1/R_J$ is the rule for tows *wrapping* a void; imposing it here pinned the design at the bound with the objective still negative.
3. **Saturating delivery reward.** Linear in E_{exit}, the delivered energy reached $6300\times$ and outweighed the guard penalty $760{:}1$ — the optimizer bought delivery by making the joint $20\times$ *busier*. A log reward fixes it.

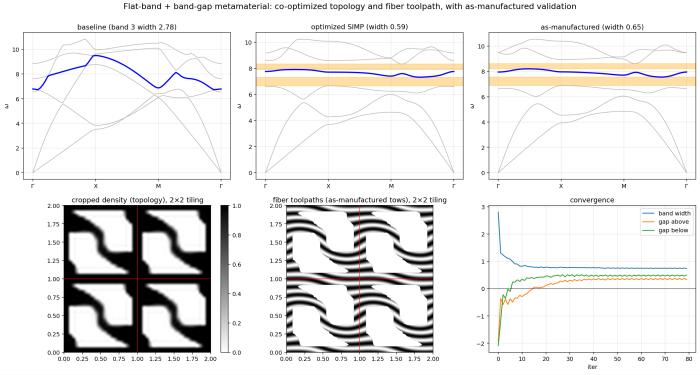
All three returned healthy headline numbers
They were caught by looking at the *field* and the *director map*, not the metric.

Periodic metamaterials

Flat band by co-designed topology and toolpath

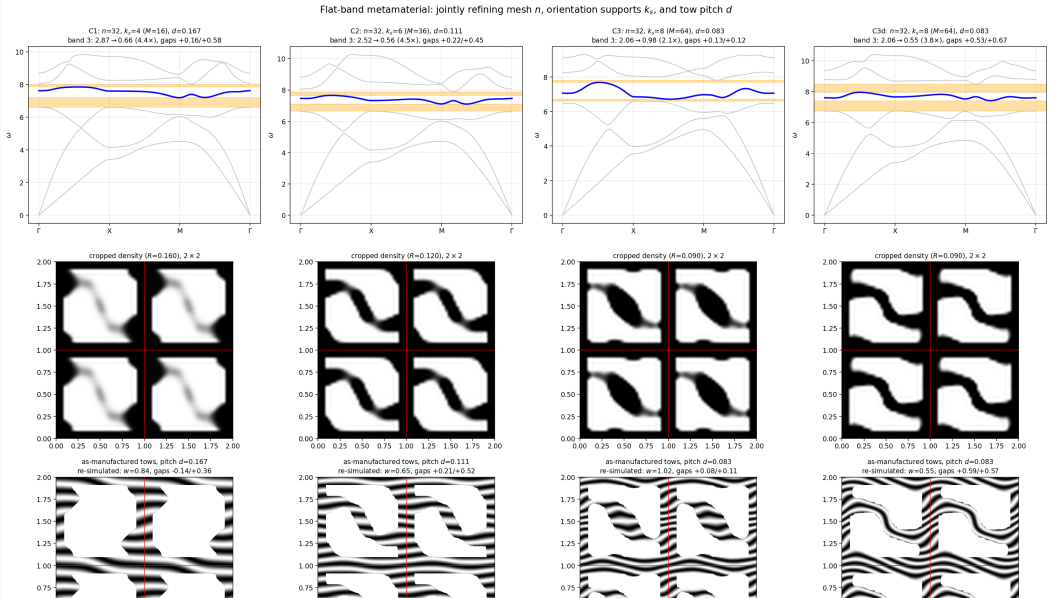
$$\mathbf{K}_r(\mathbf{k})\phi_n = \omega_n^2 \mathbf{M}_r(\mathbf{k})\phi_n, \qquad \mathbf{K}_r = \mathbf{T}^{\mathrm{H}}\mathbf{K}\mathbf{T}$$

$$J = w_{\mathrm{f}}\big(\widetilde{\max}_k \omega_b - \widetilde{\min}_k \omega_b\big) - w_{\uparrow}(\text{gap above}) - w_{\downarrow}(\text{gap below}) + c_v(\bar{z} - v^{\star})^2$$



Band 3: width 2.78 → 0.59, gaps +0.42/+0.66; survives the as-manufactured crop (0.65, +0.41/+0.67).

Does more design freedom help?



Yes — but only if the k-sampling keeps up

run	k_s (M)	N_k	w_1 (fine path)	$g_\uparrow$	$g_\downarrow$
C1	4 (16)	22	0.66	+0.16	+0.58
C2	6 (36)	22	0.56	+0.22	+0.45
C3	8 (64)	22	0.98	+0.13	+0.12
C3d	**8 (64)**	**40**	**0.55**	**+0.53**	**+0.67**

A free overfitting diagnostic

KS aggregation can only *over*-estimate a maximum. So if the width measured on a fine path *exceeds* the aggregate, the band is excursing between samples. Aggregates: 0.79, 0.70, 0.67, 0.68 against fine 0.66, 0.56, **0.98**, 0.55 — only C3, and C3 is the row that fails.

Closing

What the optimizer taught us

- **Dips in the convergence curve were the optimizer, not the physics.** A line scan is smooth; excess J scales as $(\text{step})^2$. The MMA had no globalization and a move limit $1.84\times$ the design magnitude. An adaptive trust region: $23.5\times \rightarrow 39.6\times$, zero dips.
- **Conforming meshes matter.** A staircased soft-void rim scatters on its own account; a large part of what the soft-void cloak cancelled was an artefact of the mesh.
- **Region specification is where the bugs hide.** Overlapping reward and penalty; rewarding accumulation instead of transport. Both returned confident numbers for designs doing the wrong thing.
- **MUMPS over PETSc's built-in LU:** $4.4\times$, identical answers.

Summary

Orientation-only design, one material, manufacturable toolpaths throughout:

wave lens	78× focus gain
under $\lvert\zeta\rvert \le \zeta_{\text{all}}$	12×
through-holes (conforming)	49× / 17× constrained
asymmetric holes	50× (re-solved per geometry)
elastic cloak (shell-confined)	36× scatter reduction
joint routing	698× exit, guard 0.17, conf. 97%
flat band	2.78 → 0.59, both gaps open

All of it rests on one exact statement: for an isotropic plate $\partial \mathbf{D}_f / \partial \theta \equiv \mathbf{0}$, so the identical code returns *exactly* the baseline. Anisotropy — supplied physically by the continuous-fiber toolpath — is the necessary and sufficient mechanism.